

$aba^2b = e$
 $ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$





 $\frac{S^1 \vee S^1}{\mathbb{Z} \times \mathbb{Z}}$
 $T = \mathbb{Z} \times \mathbb{Z}$
 $T \hookrightarrow \mathbb{Z} \times \mathbb{Z}$
 $\mathbb{Z} \hookrightarrow \mathbb{Z} \times \mathbb{Z}$

[illegible]

$\mathbb{R}/\mathbb{P}^2$: 0-cell: $\{v\}$
 1-cell: $\{x, y\}$
 2-cell: $\{z\}$

$S^1 \vee S^1$

$ab^{-1}b^* = 0$
 $ab = b^{-1}$

Handwritten sketches and notes on a grid background:

- A circle with a wavy line inside, two points on the circumference, and a line segment connecting them.
- A five-petaled flower shape.
- A complex, multi-lobed shape with several small circles attached to its boundary.
- A simple oval shape.
- A large, irregular, teardrop-like shape.
- Handwritten text: $\mathbb{R}^n \times \mathbb{R}^n$, $\cup \mathbb{R}^n \times \mathbb{R}^n$, $\cup (0,0,2)$, and $\mathbb{R}^n \times \mathbb{R}^n$.
- A small square with a circle inside.
- A large, irregular, teardrop-like shape.

$$Z = \frac{1}{x+y}$$

$$K_1 = \frac{1}{x+y}$$

$$K_2 = \frac{1}{x+y}$$

$$2\pi \cdot \frac{1}{x+y}$$

$\frac{1}{g} \frac{d\phi}{dt}$